NIIT | CAPSTONE PROJECT – III

Data Analytics and Automation using Python

PYTHON & LLM INTEGRATION

AI-Powered Personal Finance Tracker

Automated Transaction Management, Spending Analytics and AI-Powered Financial Insights

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PROJECT OVERVIEW

Business Problem & Key Objectives

The Problem Statement

Individuals may struggle to maintain consistent financial records, understand their spending patterns, and identify opportunities to improve their saving habits.

Key Objectives

- **Transaction Management:** Record and organise income and expenses.
- **Financial Summaries:** Calculate total income, expenses, and remaining balance.
- **Spending Analysis:** Identify category-wise expenditure and the highest-spending category.
- **AI-Powered Insights:** Generate savings suggestions and financial profile classifications.
- **Data Persistence:** Store and retrieve transaction records through Excel.

THREE-STAGE WORKFLOW

Integrated Finance Tracking and Analysis Pipeline

STAGE 1

Transaction Management

Core ledger system handling inputs, validation, and storage persistence.

- Add Income / Expenses
- View Ledger Transactions
- Financial Ledger Summary
- Excel Import and Export

Input & Processing

STAGE 2

Spending Analysis

NumPy/Pandas analytics engine evaluating expenditure distribution.

- Category-wise Breakdowns
- Identify Top Cost Drivers
- High Expense Flagging
- Net Balance Calculation

Analytical Computation

STAGE 3

AI-Powered Insights (Groq API)

LLM integration delivering AI-generated financial suggestions.

- Personalized Savings Strategy
- Potential Overspending Areas
- User Profile Classification
- Actionable Recommendations

Intelligent Output

1 Transaction Entry → 2 Data Storage → 3 Spending Analysis → 4 AI-Powered Insights

STAGE 1: TRANSACTION MANAGEMENT

Recording and Storing Financial Transactions

Transaction Menu

```
========== FINANCE TRACKER MENU ==========
1. Add Transaction
2. View Transactions
3. View Financial Summary
4. Spending by Category
5. Delete Transaction
6. Delete All Transactions
7. Import from Excel
8. Export to Excel
9. NumPy Analysis
10. Identify High Expenses
11. Financial Profile
12. AI Financial Insights
13. Exit
Enter your choice (1-13): ↑↓ for history. Search history with c-↑/c-↓
```

Successful Transaction Entry

```
Enter your choice (1-13):  1
Enter transaction ID:  T004
Enter type (Income/Expense):  Expense
Enter category:  Travel
Enter amount:  700
Enter date (DD-MM-YYYY):  13-10-2026
Transactions exported to finance_data_final.xlsx successfully.
Transaction added successfully!
```

Interactive Menu

Numbered Command Interface: Users navigate the tracker by simply entering the corresponding menu number (1–13) to trigger specific financial tasks.

Guided Input Prompts: Selecting option 1 initiates a step-by-step transaction workflow requesting Transaction ID, Type (Income/Expense), Category, Amount, and Date.

Validated Transaction Entry & Data Storage

Real-Time Data Validation: Ensures amounts are numeric, dates follow DD-MM-YYYY, and transaction IDs remain unique before saving.

Instant Excel Export: Every added record automatically updates and syncs with the underlying *finance_data_final.xlsx* storage file.

STAGE 1: TRANSACTION MANAGEMENT

Viewing Transactions and Managing Financial Records

View Transactions

Option 2

Displays all saved financial records in a clean, organized table so you can easily review your history.

```
Enter your choice (1-13):  2
```

	id	type	category	amount	date
0	T001	Income	Salary	17640	13-10-2026
1	T002	Expense	Travel	700	13-10-2026
2	T003	Expense	Food	500	13-10-2026
3	T004	Expense	Bills	5040	13-10-2026
4	T005	Expense	Shopping	1890	13-10-2026
5	T006	Expense	Food	795	13-10-2026

Financial Summary

Option 3

Instantly calculates your total income, total expenses, and current balance in real time.

```
Enter your choice (1-13):  3

========== FINANCIAL SUMMARY ==========
Total Income: ₹21000.00
Total Expenses: ₹19370.00
Remaining Balance: ₹1630.00
```

Single & Bulk Deletion

Options 5 & 6

Safely remove specific records using a Transaction ID, or reset all data securely with a confirmation step.

```
Enter your choice (1-13):  5
Enter transaction ID to delete:  T001
Transactions exported to finance_data_final.xlsx successfully.
Transaction deleted successfully.
```

Excel Backup & Sync

Options 7 & 8

Easily import or export your data with Excel spreadsheets to keep your records backed up and accessible.

```
Enter your choice (1-13):  7
Excel data imported successfully.
```

	id	type	category	amount	date
0	T002	Income	SALARY	17640	13-10-2026
1	T004	Expense	Travel	700	13-10-2026

```
Enter your choice (1-13):  8
Transactions exported to finance_data_final.xlsx successfully.
```

STAGE 2: SPENDING ANALYSIS

Category-Wise Expense Breakdown

Category-Wise Expense Breakdown

```
Enter your choice (1-13):  4

========== SPENDING BY CATEGORY ==========
category
Bills       7038
Food        1295
Others       952
Rent        7000
Shopping    1890
Travel      1195
Name: amount, dtype: int64

========== HIGHEST EXPENSE ==========
Category: Bills
Amount: ₹7038.00
```

DOMINANT CATEGORY

Bills (₹7,038)

Bills account for approximately 36.3% of total expenditure, making this the highest-spending category.

Observation: Bills and Rent together account for **₹14,038**, or approximately **72.5%** of total expenditure.

STAGE 2: SPENDING ANALYSIS

Expense Auditing and Financial Metrics

Threshold Flagging

Flags transactions > ₹2,000.

High Expense: Rent (₹7,000)
High Expense: Bills (₹5,040)

```
Enter your choice (1-13):  10


========== EXPENSE CHECK ==========
Travel - ₹700 - Normal Expense
Food - ₹500 - Normal Expense
Bills - ₹5040 - High Expense
Shopping - ₹1890 - Normal Expense
Food - ₹795 - Normal Expense
Rent - ₹7000 - High Expense
Others - ₹400 - Normal Expense
Travel - ₹495 - Normal Expense
Others - ₹200 - Normal Expense
Bills - ₹1998 - Normal Expense
Others - ₹100 - Normal Expense
Others - ₹252 - Normal Expense
```

Financial Profile Classification

"Spender"

Automatically tagged based on low savings ratio and high transaction outflow volume.

```
Enter your choice (1-13):  11

Financial Profile: Spender
```

NumPy Metrics

Total Transaction Value: ₹40,370

Average transaction amount: ₹2,691.33

Volume: 15 total transactions

```
Enter your choice (1-13):  9

========== NUMPY ANALYSIS ==========
Transaction amounts: [17640   700   500  5040  1890   795  7000   360   400   495   200  1998
   100  3000   252]
Total transaction amount: 40370
Average transaction amount: 2691.3333333333335
Unique amounts: [  100   200   252   360   400   495   500   700   795  1890  1998  3000
  5040  7000 17640]
Expenses greater than ₹2000: [5040 7000]
```

STAGE 3: AI-POWERED INSIGHTS

AI-Generated Financial Recommendations

OPTION 12

Automated Insights

Menu-Driven AI Recommendations

Analyses transaction ledger to automatically evaluate profiles and suggest cost reductions.

```
Enter your choice (1-13):  12

========== AI FINANCIAL INSIGHTS ==========
**Financial Summary**
- **Monthly Income:** ₹21,000
- **Total Expenses:** ₹19,370
- **Remaining Balance (savings):** ₹1,630 (≈ 7.8 % of income)

**Highest Spending Category**
- **Bills** – ₹7,038 (the largest single-category outlay)

**Three Practical Money-Saving Suggestions**
1. **Trim utility & subscription costs** – Review phone, internet, and
2. **Cook more at home** – Reduce food-out expenses by planning meals,
5-20 %.
3. **Limit discretionary shopping** – Set a weekly “shopping cap” (e.g.
ms.

**Classification**
- **Spender** – The surplus is modest, indicating most of the income is
move toward a more balanced or saver profile.
```

Key Benefit: Gives immediate standard money-saving strategies without typing prompts.

OPTION 13

Interactive Q&A

Custom Query & Assistance

Allows users to type any specific financial question or ask how to execute AI recommendations.

```
get_ai_financial_insights()


==================================================
     AI FINANCIAL ASSISTANT
==================================================
Ask any question.
Type 'exit' to return to menu.

You:  ↑↓ for history. Search history with c-↑/c-↓
```

```
You:  explain financial collateral

AI Advisor:
Financial collateral is an asset you pledge to a lender to secure a loan or credit.
If you fail to repay, the lender can seize and sell the collateral to recover the money.
Common types include property, vehicles, savings accounts, stocks, or equipment.
The value of the collateral usually must be higher than the loan amount to protect the lender.
Using collateral often lowers the interest rate because it reduces the lender’s risk.

You:  exit
Returning to menu...
```

Key Benefit: Provides personalized, conversational advice for any global query or follow-up step.

FINANCIAL FINDINGS

Key Findings & Actionable Recommendations

Key Finding	Interpretation	Actionable Recommendation
Bills is the highest-spending category at ₹7,038, followed by Rent at ₹7,000.	These are the two largest expense categories in the displayed dataset.	Review the Bills breakdown for recurring charges and identify any services or costs that can reasonably be reduced.
Bills and Rent account for 72.5% of total expenditure.	Spending is concentrated in these two categories.	Review recurring bills and assess housing costs as part of the overall budget.
The remaining balance is ₹1,630 from an income of ₹21,000, equivalent to approximately 7.8%.	The displayed balance represents a relatively small portion of the reported income.	Set a realistic savings target and review discretionary spending to identify opportunities to increase the remaining balance.
Discretionary categories include Shopping at ₹1,890, Food at ₹1,295, Travel at ₹1,195, and Others at ₹952.	Shopping is the largest of these categories.	Review non-essential purchases and establish a monthly shopping budget.

Conclusion

An Integrated Personal Finance Analysis Solution

The AI-Powered Personal Finance Tracker combines transaction management, category-wise spending analysis, and LLM-generated financial insights in a single Python application.

The analysis identified Bills as the highest-spending category at ₹7,038, with Bills and Rent together accounting for 72.5% of the displayed expenditure.

The financial summary showed a remaining balance of ₹1,630 from the reported income of ₹21,000. These findings provide a starting point for reviewing recurring costs, managing discretionary spending, and setting realistic savings goals.

Key Outcome

The project demonstrates how structured data analysis and AI-generated suggestions can help users better understand their financial activity and identify practical opportunities for improvement.

₹19,370

Total Expenditure Analysed

3-Stage

Transaction Management · Spending Analysis · AI Insights

NIIT CAPSTONE PROJECT III

Thank You!

Questions & Discussion

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